



Ahmed Aziz Ben Haj Hmida

Data Science Graduate (EPFL) with a Minor in Management,
Technology & Entrepreneurship | Specialized in leveraging AI and
Analytics to drive industrial R&D and Business Innovation.

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Education

EPFL – École Polytechnique Fédérale de Lausanne

M.Sc. in Data Science

Minor in Management, Technology & Entrepreneurship (MTE)

GPA: 5.3/6

Relevant Coursework: Applied Data Analysis, Data Science for Business, Finance, Microeconomics, Project Management & Risk Analysis, Value Chain Management, Economic Growth & Sustainability

Lausanne, Switzerland

Sep 2023 – Mar 2026

Technical University of Denmark (DTU)

3rd Year of Bachelor – Exchange

GPA: 5.35/6

Lyngby, Denmark

Aug 2022 – Jul 2023

EPFL – École Polytechnique Fédérale de Lausanne

B.Sc. in Communication Systems

GPA: 5.1/6

Lausanne, Switzerland

Sep 2020 – Jul 2023

Pioneer School of Nabeul

Baccalaureate in Mathematics

Nabeul, Tunisia

Sep 2018 – Jun 2019

Professional Experience

Rollomatic SA

Master's Thesis: Strategic Retrieval System for CNC Tools (Grade 6/6)

Le Landeron, Switzerland

Aug 2025 – Mar 2026

- Engineered an end-to-end similarity search system that reduced search latency by **96%**, drastically accelerating tool design cycles.
- Improved retrieval accuracy by **11%**, enabling design reuse and reducing manufacturing R&D costs.
- Integrated the system into industrial workflows, translating complex technical specifications into actionable tools.

Supervisors: Prof. Jean-Philippe Thiran - jean-philippe.thiran@epfl.ch, Emna Maghrebi - e.maghrebi@rollomatic.ch

Rollomatic SA

Internship: Data-Driven Optimization for Laser Machining

Le Landeron, Switzerland

Feb 2025 – Aug 2025

- Led the initial research workstream to identify inefficiencies in tool retrieval processes for laser machining units.
- Developed a prototype hybrid search pipeline integrating deep visual embeddings to streamline industrial workflows.
- Collaborated with R&D stakeholders to define KPIs, setting the strategic foundation for the subsequent production-level system.

Supervisor: Emna Maghrebi - e.maghrebi@rollomatic.ch

Research Projects

Discovering promising new startups through evaluating teams' potentials 🌀

EPFL ENTIC Lab

Sep 2023 – Dec 2023

- Harnessed ML methods to derive speakers' MBTI personalities and emotions during entrepreneurial discussions.
- Built predictive approaches to identify promising teams with high startup success potential.

Supervisor: Prof. Davide Bavato

Unsupervised Reverse Knowledge Distillation and Model Merging 🌀

EPFL LTS4 Lab

Aug 2024 – Feb 2025

- Transferred knowledge from compact task-specific models into large-scale architectures.
- Applied model merging techniques to improve niche-task performance while preserving broad generative capabilities.

Supervisor: Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

Technical Skills

Programming Languages: Python, Java, SQL, MATLAB, Scala

Business & Strategy: Excel, PowerPoint, Financial Modeling, Project Management, Risk Analysis, Power BI

Data Analytics & ML: PyTorch, Scikit-learn, RAG Pipelines, Agentic AI, PostgreSQL

Tools & DevOps: Git, Docker, GitHub Actions, Kubernetes, FastAPI, W&B, GCP, AWS, Streamlit

Personal & Academic Projects

LLM Agent–Driven Car Search Assistant  Apr 2026 - May 2026

- Built a Streamlit frontend where users describe a car in chat and LangGraph with LangChain and Ollama fills a structured profile and queries PostgreSQL.
- Displayed ranked listing cars with match scores plus Plotly and Matplotlib charts, generated PDFs for single searches and for comparing two or three saved car profiles, and loaded the listings into Postgres.

LLM Teaching Assistant for MCQ Answering  Feb 2024 – Jun 2024

- Fine-tuned Qwen1.5 using SFT and Direct Preference Optimization with RAG and 4-bit quantization.
- Developed a chatbot interface and evaluation suite using LLM-as-a-judge frameworks and cross-validation.

Machine Learning Operations  Jan 2023 – Mar 2023

- Deployed scalable ML pipelines using PyTorch, Docker, Kubernetes, CI/CD, GCP, AWS, MLflow, and DVC.

RL for Motor Skill Transfer  Feb 2024 – Jun 2024

EPFL Alexander Mathis Lab

- Trained expert reinforcement learning agents on Myosuite baoding ball tasks.
- Distilled motor strategies into novice agents to improve learning efficiency and performance.

Coin Detector and Counter  Feb 2024 – Jun 2024

- Used segmentation techniques and deep learning models (CNNs, Vision Transformers, ResNet50) for accurate coin recognition.

Certificates

Structuring Machine Learning Projects – DeepLearning.AI Sep 2023 - Oct 2023

Certificate in Artificial Intelligence – GOMYCODE Jun 2023 – Sep 2023

Associations

EPFL Coaching – Coach of CMS section Sep 2023 – Jun 2024

Rotary International – Member Jul 2018 – Apr 2019

Tunisian Red Crescent – Volunteer May 2018 – Apr 2019

Languages

Arabic/French: Native | English: C1 | German: A2